PROJECT REPORT

HABIT CONSISTENCY TRACKER

A DAILY HABIT MONITRING & SCORE PREDICTION SYSTEM

Course : Problem solving

Name: Saumya Dixit

Registration no. – 25BAI11460

Submission date : 24 November 2025

VITyarthi – Flipped classroom project

Abstract

The Habit Consistency Tracker is a small Python project created to help users keep a check on their everyday habits and understand how consistent they are over a period of time. The program stores daily habit entries, calculates how regularly each habit is completed, and even gives a simple prediction for next week's performance.

This project highlights core programming concepts such as modular design, functions, file handling, conditions, and basic user interaction—making it suitable for beginners who are learning Python in their first semester

Introduction

Building and maintaining habits is important, yet many people lose track because they do not monitor their progress regularly.

This project tries to provide a simple solution where the user can:

- Enter daily habit completion
- Check consistency scores
- View predicted scores for the next week

It shows how basic programming can help solve small day-to-day problems in an effective and simple manner.

Problem Definition

People often want to develop habits like studying, reading, exercising, or meditation. However, most of these habits fade away because there is no proper method to track daily progress.

Problem:

Students lack a straightforward tool to measure how consistently they follow their habits and how their performance might look in the coming days.

Solution:

Build a program that stores daily habit information, calculates consistency, and generates a prediction based on recent trends.

Objectives

- The main objectives of this project are:
- Provide an easy way for users to record their daily habit status
- Save each habit entry in a simple data file
- Calculate consistency scores
- Estimate next week's score using a basic formula
- Help users stay disciplined and track improvement
- Apply fundamental programming concepts to a real-world mini project

Requirement Analysis

- Python 3.x installed
- Any code editor (VS Code recommended)
- Standard Python libraries
- CSV file for storing data
- No third-party libraries needed

System Design

Top-Down Design

- The project is organized into three main modules:
- main.py – Handles the user menu and interactions

- .py – Used for saving and reading data from CSV
- Logic Module – Calculates consistency and prediction results

Flowchart

```
┌──────────────────────────────────────┐
│      Start Program     │
└──────────────────────┬───────────────┘
                       │
                       ▼
  ┌──────────────────────────┐
  │   Display Menu   │
  └──────────────────┬───────┘
                     │
```

Option 1: Add Daily Habit Data

Ask habit name → yes/no → Save to CSV

|

Option 2: Show Consistency Score

Read CSV → Calculate % → Display

|

Option 3: Predict Next Week Score

Apply formula → Display

|

Option 4: Exit Program

|

```
┌──────────────▼──────────────┐
|      End        |
└─────────────────────────────┘
```

Algorithm

Algorithm: Adding Habit Data

1. Start
2. Ask the user for the habit name
3. Ask if the habit was completed (yes/no)
4. Convert yes → 1 and no → 0
5. Store the data in the CSV file
6. Stop

Algorithm: Consistency Score

1. Open the CSV file

2. Count the total number of entries

3. Count completed entries

4. Calculate:

consistency = (completed / total) × 100

5. Show the result

Algorithm: Prediction

1. Calculate present consistency

2. Use the logic:

next_week ≈ consistency × (0.95 to 1.05)

3. Display predicted score

Implementation

- Files Used
- main.py: Displays menu and handles user inputs
- database.py: Manages CSV file operations
- habits.csv: Stores all habit entries
- The whole program is written using basic Python without any external libraries.

Sample Output

Habit Consistency Tracker

1. Add Daily Habit Data

2. Show Consistency Score

3. Predict Next Week Score

4. Exit

Enter option: 1

Enter habit name: reading

Did you complete it today? (yes/no): yes

Data added successfully!

Enter option: 1

Enter habit name: workout

Did you complete it today? (yes/no): no

Data added successfully!

Enter option: 2

Habit: reading → Consistency: 100%

Habit: workout → Consistency: 0%

Enter option: 3

Predicted next week score for reading: 92%

Predicted next week score for workout: 10%

Enter option: 4

Exiting... Thank you!

Conclusion

The Habit Consistency Tracker fulfills its purpose by offering a straightforward way to record habits, calculate consistency, and make simple predictions. It shows how basic programming concepts such as file handling, loops, and conditional statements can be combined to form a functional and practical application.

It is easy to use, efficient, and useful for anyone who wants to stay committed to their daily habits.